

Project 18

Local versus Global Joint-Limit Avoidance on iiwa 7

Format: Group of two students; simulation-based project.

Course foundations: Lecture 8; Exercise 4.

Motivation

Null-space motion can improve a local joint-centering objective while missing a feasible route elsewhere in the redundant IK set. Comparing that local strategy with a global feasibility chart helps distinguish algorithmic limitations from actual restrictions imposed by the robot and the task.

Task

Overall objectives. Compare local null-space joint centering with planning through a classified iiwa 7 IK chart, and explain success or failure through feasible self-motion intervals and their connections along the path.

- **Establish one common problem.** Validate the iiwa model, limits, and complete pose path. Use one short path segment that approaches a joint limit and two admissible initial configurations, ensuring both methods are compared from the same initial configuration in each trial.
- **Implement local joint centering.** Use the geometric Jacobian with a documented scaling and a projected gradient for $h(q) = \frac{1}{2} \sum_i ((q_i - q_{i,c})/r_i)^2$. Integrate small steps, correct the full-pose error, and recheck limits. Explain why quadratic centering is a preference, not a hard limit barrier.
- **Build the global comparison chart.** Enumerate the supplied fixed- q_3 IK families on path-sample versus redundant-angle axes. Preserve branch labels, plot feasible intervals and failure reasons, and distinguish full-Jacobian rank loss from reduced-chart degeneracy.
- **Plan under the same constraints.** Select a chart corridor from the same starting configuration, using joint centering and joint travel as documented costs. Validate intermediate transitions and apply the same pose, limit, and regularity tolerances used for the local method.
- **Diagnose the difference.** Overlay both trajectories on the classification charts. Compare completion, minimum limit margin, travel, and pose error for the two initial states. Repeat the local method with one smaller integration step to distinguish discretization error from failure to discover another feasible corridor.

Specifics and scope

Use one path segment, two initial configurations, and two local step sizes. Start with about 120 path samples and 120 redundancy angles and refine decisive intervals. Reuse the course IK and the chart construction from Exercise 4; a complete global optimizer is unnecessary. Equal results are acceptable when the explanation is supported by the feasibility maps.

For every chart, keep pose validity, joint limits, full-Jacobian regularity, and reduced-chart regularity as separate diagnostics. The rectangular 6×7 Jacobian has no determinant. A feasible pixel is not a validated edge: re-solve intermediate prescribed poses and check joint continuity within limits. If collision checking is unavailable, state that feasibility is kinematic only; a clearance proxy is not a full collision test.

Expected outcome

Understand the difference between instantaneous null-space freedom and global feasible connectivity. Deliver classified redundancy charts with trajectory overlays, matched comparisons, and an explanation of whether observed limitations arise from the strategy, numerical steps, or constraints.

Submit reproducible code, model parameters and test data, the required plots, a concise 5–6-page report, and a short simulation demonstration. Explain both successful cases and failures; conclusions must follow the numerical and geometric evidence rather than an expected outcome.

Resources

The KUKA LBR iiwa 7 URDF is `lectures_main/assets/models/iiwa7/iiwa7.urdf`; STL meshes are in that directory. The accompanying `iiwa7_free_joints.urdf` is a mathematical continuation model; apply physical limits separately. The course analytical IK model will be supplied; reuse it rather than deriving a general 7R solver.

The course pose path is `lectures_main/assets/data/path/kuka_iiwa7_rectangle_workspace_path_500_quaternion.csv`. Its quaternion columns are `qw,qx,qy,qz`; normalize them and make their signs consistent before interpolation. The reference table `lectures_main/assets/data/null/iiwa7_IK_q3_030deg.csv` checks fixed- q_3 solutions only; it does not replace evaluation at other angles. Use the null-space derivation and feasibility-map demonstrations in Lecture 8. Choose a short segment or documented placement of the course path; a suitable joint-limit example can be supplied to keep scenario construction bounded.

Use the lecture website to verify D–H parameters, visualize IKs and singularities, and simulate paths. All listed paths are relative to the repository root. Start the website following `lectures_main/README.md`.

Workload and collaboration

Budget approximately 60 hours per student (120 person-hours per pair) for this 2-ECTS project, following EPFL's 30-hours-per-credit convention.¹ Deduct any lectures or exercises included in those credits. Allocate roughly 20 team-hours to model validation, 50 to the core work, 30 to experiments, and 20 to reporting. Reuse the specified IK and simulations. Share validation and interpretation even if modelling and implementation are divided between students. Hardware execution is outside scope.

¹EPFL: Design a Course, workload guidance.